Linux OS

Day 1: System Basics

Files path : davinci0x01_lvl0_Tasks\Skill_4_Linux\Day 1

1.Display the Linux distribution name and version > 1.1.sh

- Used lsb_release -a to display distribution name/version when available

- If lsb_release is not installed, used cat /etc/os-release as a fallback

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 1]
└─$ ./1.1.sh
No LSB modules are available.
Distributor ID: Kali
Description:    Kali GNU/Linux Rolling
Release:        2025.4
Codename:       kali-rolling
```

2.Print current username and UID > 1.2.sh

- Used whoami to print the currently logged-in username

- Used id -u to print the user ID (UID) of the current user

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 1]
└─$ ./1.2.sh
Current User: kali
User ID (UID): 1000
```

3.Show the system's hostname and local IP address > 1.3.sh

- Used hostname to retrieve the system hostname

- Used hostname -I and awk to display the primary local IP address

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 1]
└─$ ./1.3.sh
Hostname: kali
Local IP: 10.0.2.15
```

4.List all environment variables > 1.4.sh

- Used `printenv` command to display all environment variables

- The script outputs variable names and their corresponding values.

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 1]
└─$ ./1.4.sh
XDG_DATA_DIRS=/usr/share/xfce4:/usr/local/share/:/usr/share/:/usr/share
GDK_BACKEND=x11
PATH=/home/kali/.local/bin:/usr/local/sbin:/usr/sbin:/sbin:/usr/local/bin:/usr/bin:/bin:/usr/local/games:/usr/games:/home/kali/.dotnet/tool
s
GDMSESSION=lightdm-xsession
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
OLDPWD=/home/kali/Downloads/Skill_4_Linux
TERM_PROGRAM=vscode
```

5.Detect whether the current system is running Linux > 1.5.sh

- Used `uname` command to retrieve the operating system name.

- Used a conditional statement to check if the result equals "Linux"

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 1]
└─$ ./1.5.sh
The system is running Linux.
```

6.Display the system uptime in a readable format > 1.6.sh

- Used uptime -p command to display system uptime

- The output shows how long the system has been running since the last boot

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 1]
└─$ ./1.6.sh
up 2 minutes
```

7.Show the CPU model and count the number of cores > 1.7.sh

- Used lscpu with `grep` to display the CPU model name

- Used nproc to retrieve the number of CPU cores

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 1]
└─$ ./1.7.sh
Model name: Intel(R) Core(TM) i5-6440HQ CPU @ 2.60GHz
Number of CPU Cores: 3
```

8.Display total and available RAM in a human-readable format > 1.8.sh

- Used free -h to display RAM usage in a human-readable format

- The output shows total, used, free, and available memory

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 1]
└─$ ./1.8.sh
               total        used        free      shared  buff/cache   available
Mem:           3.8Gi       1.4Gi       1.4Gi        38Mi       1.3Gi       2.5Gi
Swap:          953Mi          0B       953Mi
```

9.Determine whether the system is 32-bit or 64-bit > 1.9.sh

- Used uname -m to retrieve system architecture

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 1]
└─$ ./1.9.sh
x86_64
```

10.Monitor CPU and memory usage live in the terminal > 1.10.sh

- Used top command to monitor CPU and memory usage in real time.

- The tool displays running processes and system resource consumption dynamically

Screenshot:

```
top - 19:58:00 up 20 min,  1 user,  load average: 0.33, 0.31, 0.23
Tasks: 204 total,   1 running, 203 sleeping,   0 stopped,   0 zombie
%Cpu(s): 13.5 us,  4.3 sy,  0.0 ni, 82.1 id,  0.0 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem :   3919.7 total,   1308.5 free,   1469.1 used,   1415.8 buff/cache
MiB Swap:    953.7 total,    953.7 free,      0.0 used.   2450.6 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1918 kali      20   0   32.6g 214832 139780 S  26.2   5.4   0:27.03 code
   1990 kali      20   0 1395.3g 327788 143080 S  13.3   8.2   1:02.54 code
    744 root      20   0  434940 131360  83608 S   6.6   3.3   0:27.15 Xorg
   1868 kali      20   0 1393.8g 193644 139652 S   4.0   4.8   0:12.43 code
   1334 kali      20   0  600448 142536  93692 S   1.3   3.6   0:05.83 xfwm4
   1393 kali      20   0  303836  56044  21524 S   0.7   1.4   0:03.42 wrapper-2.0
   1395 kali      20   0  272844  29640  22480 S   0.7   0.7   0:03.06 wrapper-2.0
   2120 kali      20   0 1409.6g 116980  78556 S   0.7   2.9   0:04.10 code
     15 root      20   0       0      0      0 I   0.3   0.0   0:00.40 rcu_preempt
     33 root      20   0       0      0      0 I   0.3   0.0   0:00.04 kworker/u13:0-kvfree_rcu_reclaim
    488 message+  20   0   10212   6660   4324 S   0.3   0.2   0:01.38 dbus-daemon
   1398 kali      20   0  285880  38676  30760 S   0.3   1.0   0:00.56 wrapper-2.0
  12277 kali      20   0   10696   5924   3748 R   0.3   0.1   0:00.10 top
      1 root      20   0   24616  15184  11328 S   0.0   0.4   0:01.58 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.01 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
```

Day 2: Files & Permissions

Files path : davinci0x01_lvl0_Tasks\Skill_4_Linux\Day 2

1.Create a file and write the text "Hello Linux" into it > 2.1.sh

- Used echo with redirection operator > to create a new file

- The script writes the text "Hello Linux" inside hello.txt

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 2]
└─$ ./2.1.sh
File created successfully.

┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 2]
└─$ cat hello.txt
Hello Linux
```

2.Check the size of a specific file in bytes > 2.2.sh

- Used stat -c%s to retrieve file size in bytes

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 2]
└─$ ./2.2.sh
File Size in Bytes: 12
```

3.List all files (including hidden) in a directory with their permissions and sizes > 2.3.sh

- Used ls -lah to list all files including hidden files

- The command displays permissions, owner, file size, and modification date

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 2]
└─$ ./2.3.sh
total 24K
drwxrwxr-x 2 kali kali 4.0K Feb 26 20:10 .
drwxrwxr-x 5 kali kali 4.0K Feb 20 22:37 ..
-rwxrwxr-x 1 kali kali   78 Feb 26 20:01 2.1.sh
-rwxrwxr-x 1 kali kali   62 Feb 26 20:08 2.2.sh
-rwxrwxr-x 1 kali kali   20 Feb 26 20:11 2.3.sh
-rw-rw-r-- 1 kali kali   12 Feb 26 20:02 hello.txt
```

4.Change the permissions of a script to make it executable > 2.4.sh

- Used `chmod +x` to add execute permission to the script file

- This command is used before executing any .sh script :)

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 2]
└─$ ./2.4.sh
chmod +x example.sh
 :)
```

5.Recursively change the owner of all files in a directory to a specific user > 2.5.sh

- Created a test directory and file to demonstrate ownership changes

- Created a temporary user, then used chown -R user:user directory to change ownership recursively

- Verified the result using ls -lah then removed the test user and directory for cleanup

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 2]
└─$ ./2.5.sh
##### Create test_folder and file in it #####
Done
##### Create User #####
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for test
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n]
Ownership changed successfully.
total 8.0K
drwxrwxr-x 2 test test 4.0K Feb 26 20:42 .
drwxrwxr-x 3 kali kali 4.0K Feb 26 20:42 ..
-rw-rw-r-- 1 test test    0 Feb 26 20:42 test_file.txt
##### Delete user #####
userdel: test mail spool (/var/mail/test) not found
User Deleted
##### Delete folder #####
Folder deleted
```

6.Find all files larger than 100MB on the system > 2.6.sh

- Used find to search the entire filesystem for files larger than 100MB

- Used 2>/dev/null to suppress permission errors while scanning system directories

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 2]
└─$ ./2.6.sh
/media/sf_Files/en_windows_7_ultimate_with_sp1_x64_dvd_u_677332.iso
/media/sf_Files/kali-linux-2025.4-virtualbox-amd64/kali-linux-2025.4-virtualbox-amd64.vdi
/media/sf_Files/kali-linux-2025.4-virtualbox-amd64.7z
/media/sf_Files/ubuntu-18.04.5-live-server-amd64.iso
/boot/initrd.img-6.18.5+kali-amd64
/boot/initrd.img-6.16.8+kali-amd64
/proc/kcore
/home/kali/.local/share/Trash/files/Image.ad1
/home/kali/.local/share/Trash/files/code_1.109.5-1771531656_amd64.deb
/home/kali/.BurpSuite/burpbrowser/144.0.7559.59/chrome
/var/lib/docker/overlay2/5715c581983a80687804ece2a2c229d5d2c090fdd2a3278946d145969028c086/diff/usr/local/bin/node
```

7.Create a symbolic link to an existing file > 2.7.sh

- Used ln -s to create a symbolic link to an existing file

- Verified the link creation using ls -l which shows the link arrow ->

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 2]
└─$ ./2.7.sh
Symbolic link created successfully.
lrwxrwxrwx 1 kali kali 9 Feb 26 20:52 hello_link.txt -> hello.txt
```

8.Display the owner and permissions of a file > 2.8.sh

- Used ls -l to display file permissions and ownership details

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 2]
└─$ ./2.8.sh
-rw-rw-r-- 1 kali kali 12 Feb 26 20:02 hello.txt
```

9.Compare the contents of two text files > 2.9.sh

- Used diff command to compare two text files line by line

- The command displays the differences between the two files if they are not identical

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 2]
└─$ ./2.9.sh
##### hello.txt to file.txt #####
1c1
< Hello Linux
---
> Just txt File :)
##### hello.txt to test-file.txt #####
```

10.Show the disk usage of each subdirectory inside /home > 2.10.sh

- Used du -h --max-depth=1 /home to display disk usage of each direct subdirectory inside /home

- The --max-depth=1 option ensures that only top-level subdirectories are displayed instead of recursive output

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 2]
└─$ ./2.10.sh
2.7G    /home/kali
2.7G    /home
```

Day 3: Processes & Services

Files path : davinci0x01_lvl0_Tasks\Skill_4_Linux\Day 3

1.List all currently running processes > 3.1.sh

- Used ps aux to list all running processes in the system

- The command displays process owner, PID, CPU usage, memory usage, and command name

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 3]
└─$ ./3.1.sh
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  1.0  0.3  24576 15576 ?        Ss   21:11   0:01 /sbin/init splash
root           2  0.0  0.0      0     0 ?        S    21:11   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    21:11   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   21:11   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   21:11   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   21:11   0:00 [kworker/R-kvfree_rcu_reclaim]
root           7  0.0  0.0      0     0 ?        I<   21:11   0:00 [kworker/R-slub_flushwq]
root           8  0.0  0.0      0     0 ?        I<   21:11   0:00 [kworker/R-netns]
root           9  0.0  0.0      0     0 ?        I    21:11   0:00 [kworker/0:0-events]
root          10  0.0  0.0      0     0 ?        I<   21:11   0:00 [kworker/0:0H-events_highpri]
root          11  0.0  0.0      0     0 ?        I    21:11   0:00 [kworker/0:1-events]
```

2.Find the process ID (PID) of a running program > 3.2.sh

- Used service apache2 start to ensure the service is running

- Used pgrep apache2 to retrieve all process IDs associated with the service

- Displayed PIDs in a single formatted line using tr

- Stopped the service after verification using service apache2 stop

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 3]
└─$ ./3.2.sh
PIDs: 13803 13806 13807 13808 13809 13811
```

3.Kill a process by its name > 3.3.sh

- Used service apache2 start to ensure the process is running

- Used pgrep to display all PIDs associated with apache2

- Used pkill apache2 to terminate all processes with that name

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 3]
└─$ ./3.3.sh
##### Starting apache2 #####
##### Checking process IDS ######
PIDs: 23112 23115 23116 23117 23118 23119
##### Killing apache2 process #####
[sudo] password for kali:
apache2 has been successfully terminated
```

4.Display the memory usage of a specific process > 3.4.sh

- Used ps -o pid,%mem,cmd -C process_name to display memory usage of a specific process

- Started the service before checking to ensure the process exists

- Displayed PID and memory percentage used by the process

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 3]
└─$ ./3.4.sh
##### Starting apache2 #####
##### Memory Usage of apache2 #####
    PID %MEM CMD
  26726  0.6 /usr/sbin/apache2 -k start -DFOREGROUND
  26729  0.3 /usr/sbin/apache2 -k start -DFOREGROUND
  26730  0.3 /usr/sbin/apache2 -k start -DFOREGROUND
  26731  0.3 /usr/sbin/apache2 -k start -DFOREGROUND
  26732  0.3 /usr/sbin/apache2 -k start -DFOREGROUND
  26733  0.3 /usr/sbin/apache2 -k start -DFOREGROUND
##### Stopping apache2 #####
```

5.Show the top five processes by CPU usage > 3.5.sh

- Used ps aux --sort=-%cpu to sort processes by CPU usage in descending order

- Used head -n 6 to display the top five processes along with the header row

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 3]
└─$ ./3.5.sh
##### Top 5 Processes by CPU Usage #####
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
kali       28905  100  0.1   9812  4472 pts/0    R+   21:57   0:00 ps aux --sort=-%cpu
kali        2030  4.4  8.8 1463039620 354436 ?   Sl   21:12   2:01 /usr/share/code/code --type=zygote
kali        1984  2.6  5.2 50925904 209476 ?     Sl   21:12   1:12 /usr/share/code/code --type=zygote --no-zygote-sandbox
root         742  2.4  3.3 443168 133040 tty7    Rsl+ 21:12   1:06 /usr/lib/xorg/Xorg :0 -seat seat0 -auth /var/run/lightdm/root/:0 -nolist
en tcp vt7 -novtswitch
root           1  1.4  0.3  25020 15636 ?        Ss   21:11   0:39 /sbin/init splash
```

6.Start a systemd service using systemctl > 3.6.sh

- Used systemctl start to start the specified service

- Used pgrep to verify that the service is running by checking its process IDs

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 3]
└─$ ./3.6.sh
##### Starting apache2 using systemctl #####
[sudo] password for kali:
##### Checking PID #####
PIDs: 32274 32277 32278 32279 32280 32281
```

7.Check whether a specific service is enabled at boot > 3.7.sh

- Used systemctl is-enabled to check whether NetworkManager starts automatically at boot

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 3]
└─$ ./3.7.sh
##### Checking if NetworkManager is enabled at boot #####
NetworkManager is enabled at boot.

┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 3]
└─$ ./3.7.sh
##### Checking if apache2 is enabled at boot #####
apache2 is disabled at boot.
```

8.Restart a specific service > 3.8.sh

- Used systemctl restart to restart the NetworkManager service

- Verified successful restart by retrieving the service PIDs using pgrep

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 3]
└─$ ./3.8.sh
##### Restarting NetworkManager #####
##### Checking PID after restart #####
PIDs: 37941
```

9.View logs for a service using journalctl > 3.9.sh

- Used journalctl -u service_name to retrieve service logs

- Limited the output to the last 20 log entries using -n 20

- Used --no-pager to display logs directly in the terminal

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 3]
└─$ ./3.9.sh
##### Showing last 20 logs for apache2 #####
Mar 01 21:53:35 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...
Mar 01 21:53:35 kali apachectl[26188]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0
.1.1. Set the 'ServerName' directive globally to suppress this message
Mar 01 21:53:35 kali systemd[1]: Started apache2.service - The Apache HTTP Server.
Mar 01 21:53:41 kali systemd[1]: Stopping apache2.service - The Apache HTTP Server...
Mar 01 21:53:41 kali apachectl[26293]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0
.1.1. Set the 'ServerName' directive globally to suppress this message
Mar 01 21:53:41 kali systemd[1]: apache2.service: Deactivated successfully.
Mar 01 21:53:41 kali systemd[1]: Stopped apache2.service - The Apache HTTP Server.
Mar 01 21:54:18 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...
Mar 01 21:54:18 kali apachectl[26726]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0
```

10.Monitor live process activity using a terminal tool > 3.10.sh

- Used top command to monitor running processes in real time

- The tool displays CPU usage, memory usage, and active processes dynamically

Screenshot:

```
┌──(kali㉿kali)-[~/Downloads/Skill_4_Linux/Day 3]
└─$ ./3.10.sh
##### Launching live process monitor (top) #####
top - 22:21:36 up  1:09,  1 user,  load average: 0.29, 0.29, 0.29
Tasks: 208 total,   1 running, 207 sleeping,   0 stopped,   0 zombie
%Cpu(s): 13.3 us,  6.3 sy,  0.0 ni, 79.9 id,  0.1 wa,  0.0 hi,  0.4 si,  0.0 st
MiB Mem :   3919.7 total,    761.0 free,   1750.5 used,   1725.9 buff/cache
MiB Swap:    953.7 total,    953.7 free,      0.0 used.   2169.2 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   2030 kali      20   0 1395.3g 375552 145812 S  21.5   9.4   3:09.33 code
   1984 kali      20   0   48.6g 215608 140712 S   9.9   5.4   1:45.32 code
    742 root      20   0  455216 140700  85660 S   9.6   3.5   1:35.22 Xorg
   1934 kali      20   0 1393.8g 208076 143332 S   6.9   5.2   0:32.57 code
   1332 kali      20   0  604544 143016  94204 S   1.7   3.6   0:17.89 xfwm4
   2075 kali      20   0 1409.6g 115604  79108 S   0.7   2.9   0:11.81 code
   1263 kali      20   0  215580   3384   2964 S   0.3   0.1   0:06.48 VBoxClient
   1391 kali      20   0  304180  58788  24436 S   0.3   1.5   0:11.65 wrapper-2.0
```